8086 assembly language programming

LAB-8

Structure of an Assembly Language Program

```
. model small        ; Select a memory model.
. stack              ; Define the stack size
.data                ; Variable and array declarations, Declare variables at this level
. code
main proc            ; Write the program main code at this level main
endp
; Other Procedures
                     ; Always organize your program into procedures
end main             ; To mark the end of the so
```

The Model directive:

The model directive specifies the total amount of memory the program would take. In other words, it gives information on how much memory the assembler would allocate for the program. This depends on the size of the data and the size of the program or code.

Segment directives:

Segments are declared using directives. The following directives are used to specify the following segments:

- stack
- data
- code

Stack Segment:

- Used to set aside storage for the stack
- Stack addresses are computed as offsets into this segment
- Use: stack followed by a value that indicates the size of the stack

Data Segments:

- Used to set aside storage for variables.
- Constants are defined within this segment in the program source.
- Variable addresses are computed as offsets from the start of this segment
- Use: data followed by declarations of variables or definitions of constants.

Code Segment: The code segment contains executable instructions macros and calls to procedures. Use: code followed by a sequence of program statements

1. WAP to add two 16-bit numbers.

PROGRAM:

```
.MODEL SMALL
.STACK 100H
.DATA
        NUM1 DW 1234H
        NUM2 DW 4321H
        SUM    DW 0000H
.CODE
        MAIN PROC
        MOV AX, @DATA
        MOV DS, AX
        MOV AX, NUM1
        ADD AX, NUM2
        MOV SUM, AX
        MOV AH, 4CH
        INT 21H
        MAIN ENDP
END MAIN
```

Output:

```
NUM1  = 1234
NUM2  = 4321
SUM    = 5555
```

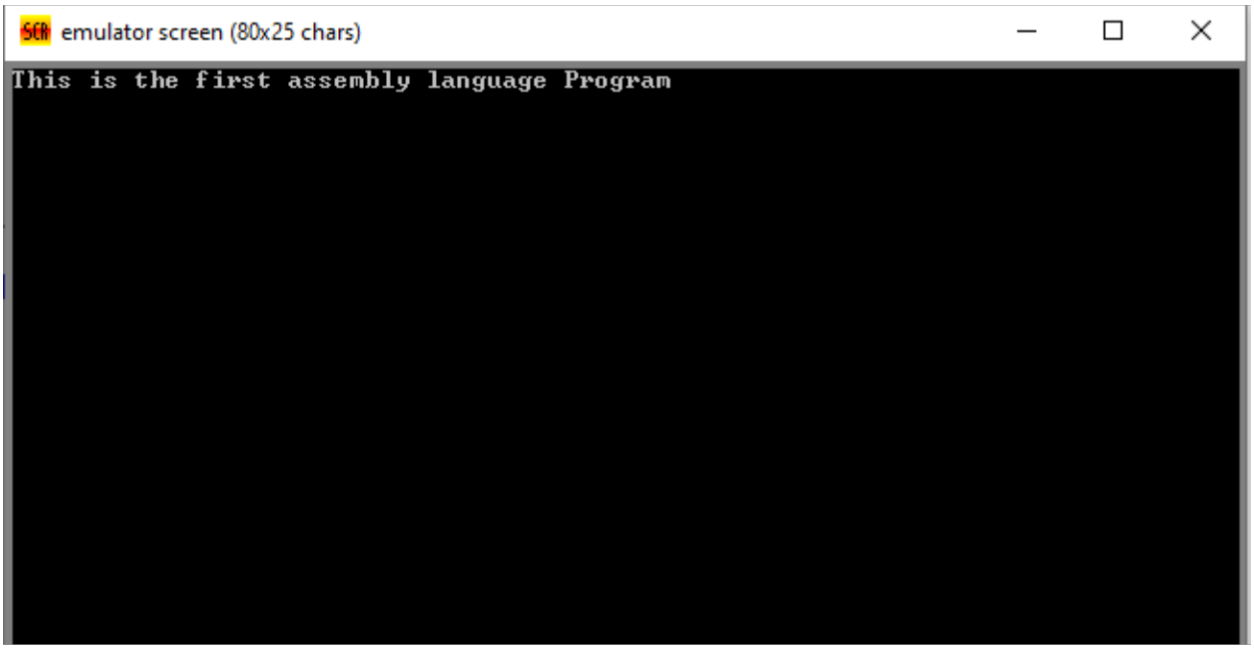
2. Experiment Name: Write a program in 8086 to display the following string. "This is the first assembly language program."

Program:

```
. model small;              small memory model required
. stack 100h;               set stack size
.data
String db "This is the first assembly language Program", '$'
. code                      ; indicates code segment starts here
Main proc                   ; indicates a procedure called main starts here
Mov ax, @data
Mov ds, ax
Mov ah, 09h
Mov dx, offset string
```

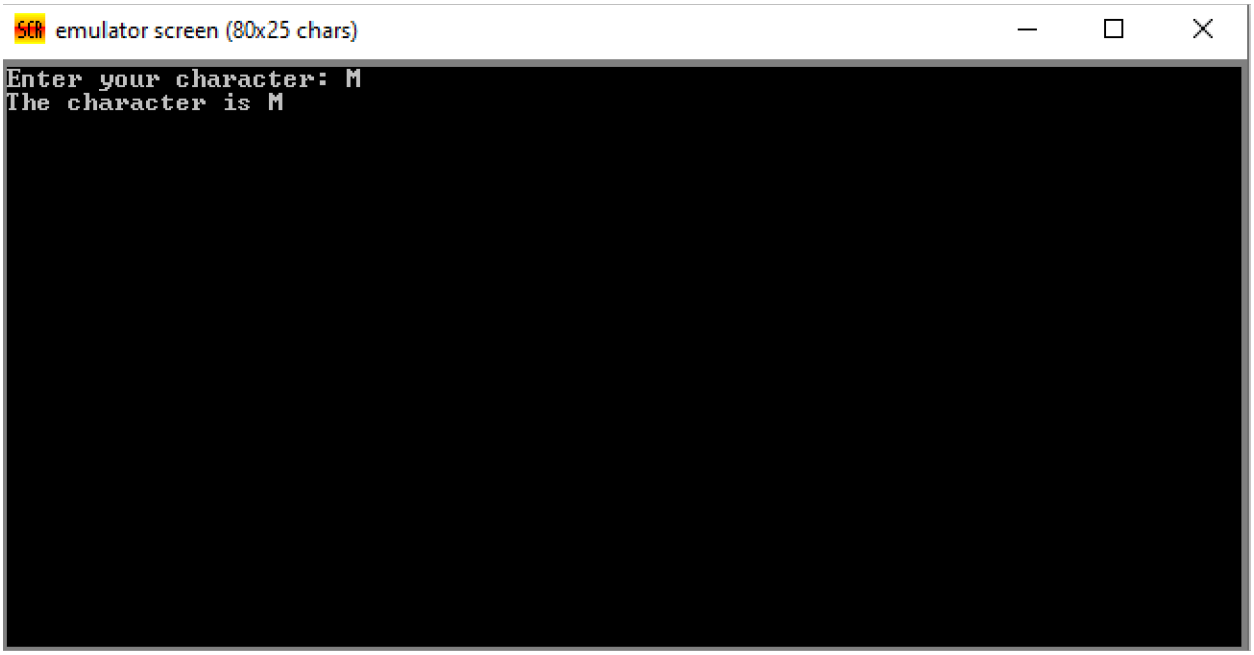
```
Int 21h
Mov ax, 4C00h                    ; exit the program and return to DOS
Int 21h
Main endp                        ; indicates that the procedure is finished.
End main                         ; tells the assembler that the program is finished.
```

Output:



3. Program to display the character Enter by user.

```
. model small
.data
Msg1 db "Enter your character: $"
Msg2 db 10,13, "The character is $"
.code
Start:
Mov ax,@data
Mov ds, ax
Mov dx, offset msg1
Mov ah, 09h
Int 21h
```

```
Mov ah,1h
Int 21h
mov bl,al
Mov dx, offset msg2
Mov ah, 09h
Int 21h
Mov dl, bl
Mov ah, 02h
Int 21h
Mov ah, 4ch
```

Output: